

STAT184 Final Project

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1 Introduction

Basketball is a high-intensity sport that relies heavily on physical characteristics. Most notably, height has always been seen as a leading factor, but how do other physical characteristics predict a basketball players performance. Wingspan, the measure of distance from fingertip to fingertip with arms fully extended, has been widely regarded as a critical role in defensive performance. A longer wingspan enables players to contest shots at a higher point, disrupt passing lanes, and generate turnovers more effectively.

In this analysis we conduct an exploratory data analysis of NBA player's physical measurements along with their defensive statistics. Our goal is to identify whether and how physical characteristics of basketball players can be used to predict their performance on the court. Our three research questions we will answer through our analysis are: 1. How do physical characteristics vary by position? 2. How does wingspan advantage affect performance? 3. Is there a clear difference in performance with wingspan advantage?

2 Data Provenance

The NBA Wingspan & Performance (Combined Dataset) was obtained from the SCORE Sports Data Repository. The dataset aggregates biometric measurements and performance statistics for NBA players from official and publicly available basketball data sources. The repository curates and combines these sources into a structured dataset for sports analytics research.

The original data creators are official stat trackers from the NBA to measure performance and NBA teams and officials to measure the biometric values. The data is maintained by the SCORE repository. The data contains both quantitative and qualitative statistics including identifiers such as players' names, teams, and season. Some quantitative statistics include the biometric measurements and performance statistics. The data was collected across multiple NBA seasons, with the biometric measurements typically taken at the combine. The data was collected physically at NBA events, such as games and combines, and digitally at official NBA databases and public sports stats platforms. The purpose of this data is to track and evaluate players physical biometrics as well as their performance, and SCORE compiles data for research and analytic purposes.

This dataset was not used for any projects in class, contains real-world data, contains rich information for exploration, and is publicly available for educational and research purposes. This dataset satisfies the FAIR principles because it is find able, available through SCORE repository, accessible, publicly downloadable, inter operable, structured format, and reusable. The dataset also satisfies most CARE principles because it has collective benefit, supports sports analytics and performance research, responsibility, we interpret this data responsibly, and with minimal ethical concern. Because the NBA ultimately owns the data with limited user control, this does not completely satisfy the CARE principle of authority to control.

3 Physical Analysis Across Positions (Sean)

In this section we will explore how biometric measurements of NBA players vary across position. This is useful in understanding the importance physical characteristics of a player have in their role and duty. By grouping the biometric measurements by position, we can see clear trends in positions that can relate to performance. In the table below, we analyze three different biometric measurements by position: Height (in), Wingspan (in), and Wingspan Advantage (in). We calculate their mean and standard deviation for each of these across all positions. It shows how Centers and Power Forwards on average have a higher height and wingspan.

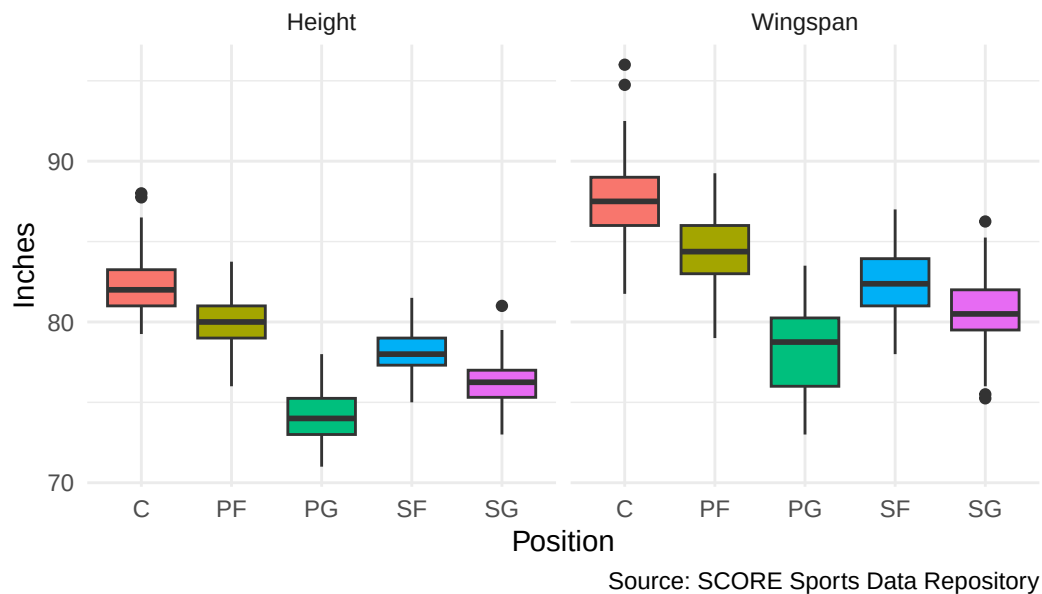
Table 1: Physical Characteristics by Position

Position	Height (in)		Wingspan (in)		Wingspan Advantage (in)	
	Mean	SD	Mean	SD	Mean	SD
C	82.31	1.79	87.38	2.64	5.08	2.29
PF	80.02	1.66	84.42	1.95	4.40	2.10
PG	74.05	1.64	78.33	2.51	4.28	2.09
SF	78.26	1.32	82.49	2.09	4.24	2.31

SG	76.23	1.37	80.58	2.22	4.35	2.20
----	-------	------	-------	------	------	------

The plot below shows boxplots for each position with respect to both height and wingspan. From this we can clearly see trends across positions. Centers clearly have the largest height and wingspan spread, followed closely by both forward positions. This may be insightful later into understanding the defensive contributions these positions have. Both guard positions have a smaller height and wingspan spread. Point guards specifically have the lowest height and wingspan spread, which makes sense because they are generally the most skillful players on their team. These boxplots highlight the importance of height and wingspan in basketball.

Physical Characteristics by Position



Alt text: Boxplots comparing NBA player height and wingspan by position.

Long Description: Two boxplot panels show height and wingspan (in) for NBA positions (C, PF, PG, SF, SG). Centers have the highest medians in both height and wingspan, followed by power forwards and small forwards. Shooting guards are slightly smaller, and point guards are the smallest. Distributions overlap but show a clear size trend by position.

4 Wingspan Advantage vs Statistical Performance (Charlie)

This section explores the Wingspan Advantage of NBA players, which takes their wingspan in inches and subtracts their height from it. I attempted to explore how much Wingspan advantage has an impact on the statistical performance of the players, and if greater wingspan advantage leads players to have higher overall statistics. I also mainly focused on defensive areas such as rebounds, blocks, and steals because from scanning the dataset they seemed to be the most variety in those areas.

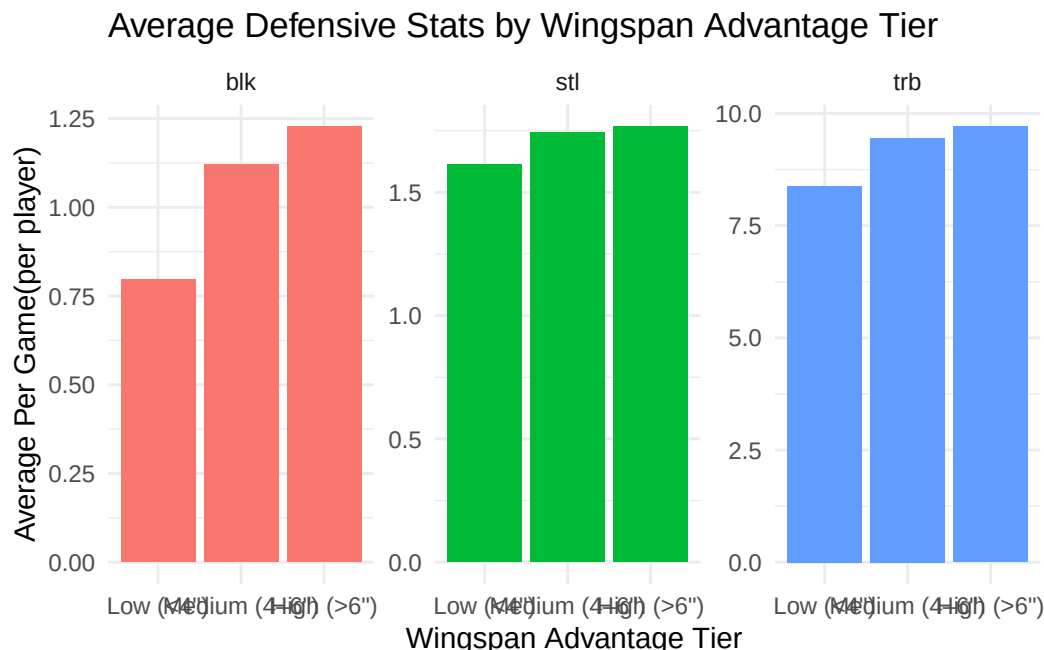
This analysis follows an EDA paradigm (exploratory data analysis), because we are trying to indentify patterns in the data and not necesarrily look at a traditional hypothesis. In the coding, it is a Tidyverse perspective in R. This is because using packages such as 'dplyr', 'tidyr' and ggplot2 are the best for summarizing, analysing and visualizing the data.

For this dataset, each case represents an individual and unique NBA player. Each row represents one player and includes both performance stats such as points and rebounds, as well as physical attributes such as height and wingspan. The main case attributes used in my section are points, steals, blocks, rebounds, and wingspan advantage(defined earlier as the difference between wingspan and height in inches.). The stats are measured as per 100 possession performance.

Table 2

wingspan_tier	avg_blk	avg_trb	avg_stl	avg_pts
High (>6)	1.2276190	9.709524	1.766667	21.78952
Low (<4)	0.7958824	8.381177	1.613529	21.74235
Medium (4-6)	1.1209790	9.454546	1.743357	21.17622

This table shows us the average statistics for the NBA players, as grouped by wingspan tier. I categorized the players based on if they had a relatively low wingspan advantage (<4 inches), a medium one (4-6 inches) or a high one (>6 inches). When looking at this table we are able to analyze if a larger wingspan advantage leads players to perform better in certain categories, regardless of position. Overall there was a difference with the players with a high wingspan advantage performing better in every category, but it was the most noticable in the gap between low and medium when looking at blocks and rebounds which are more defensive statistics. We also must classify these results as descriptive though, as it cannot be proven that wingspan directly causes better statistical performance especially defensively due to other influences such as team role, team success, playing time, and position also may cause these differences.



This visualization shows average defensive stats of players across the three wingspan advantage tiers I created. It shows the average of each stat across each of the three wingspan groups so that comparisons can be made.

Overall, the results were as I expected as there was a clear correlation between wingspan advantages and higher blocks and rebounds. Within context of basketball this makes a lot of sense, as longer arms allows for longer reach to grab rebounds, as well as block opponent shots. When looking at steals, the pattern isn't as strong because that is a more perimeter defensive stat, and depends more on speed and agility more than length, even though wingspan is still an advantage.

5 Player-Level Wingspan Extremes and Performance (Suhas)

For my part, I looked at players with the highest and lowest wingspan advantage. Wingspan advantage is the difference between a player's wingspan and height. I wanted to see if players with a bigger wingspan advantage also had better stats, mainly blocks and rebounds.

This is exploratory because I am looking for patterns, not trying to prove wingspan causes better performance. Each row is one NBA player. The main variables I used are position, height, wingspan advantage, points, blocks, rebounds, and steals.

The table below compares the 10 lowest and 10 highest players by wingspan advantage. This helps show the actual players in each group. The higher group seems to have stronger defensive stats overall, but it is not true for every player.

Table 3: Players with the Highest and Lowest Wingspan Advantage

Group	Player	Position	Height (in)	Wingspan Advantage (in)	PTS
Lowest Wingspan Advantage	Svi Mykhailiuk	SG	78.50	-1.50	21.1
Lowest Wingspan Advantage	Tyler Herro	SG	76.50	-1.25	33.7
Lowest Wingspan Advantage	Kelly Olynyk	C	82.75	-1.00	20.8
Lowest Wingspan Advantage	Kyle Kuzma	PF	80.00	-1.00	23.9
Lowest Wingspan Advantage	Desmond Bane	SG	77.00	-0.75	28.0
Lowest Wingspan Advantage	Josh Giddey	SF	80.00	-0.50	22.5
Lowest Wingspan Advantage	Deni Avdija	SF	80.75	-0.25	27.4
Lowest Wingspan Advantage	Kyle Filipowski	PF	82.75	-0.25	21.8
Lowest Wingspan Advantage	Mason Plumlee	C	83.25	-0.25	12.7
Lowest Wingspan Advantage	Ty Jerome	PG	76.25	-0.25	30.3
Highest Wingspan Advantage	Talen Horton-Tucker	SG	74.50	10.75	24.4
Highest Wingspan Advantage	Jalen Williams	SG	76.50	9.75	32.0
Highest Wingspan Advantage	Isaiah Stewart	C	79.25	9.50	14.6
Highest Wingspan Advantage	Robert Williams	C	80.00	9.50	16.1
Highest Wingspan Advantage	Bilal Coulibaly	SF	78.00	9.00	17.8
Highest Wingspan Advantage	Josh Okogie	SF	75.00	9.00	22.3
Highest Wingspan Advantage	Kawhi Leonard	SF	78.00	9.00	33.2

Highest Wingspan Advantage	Donovan Mitchell	SG	73.25	8.75	36.7
Highest Wingspan Advantage	Andre Drummond	C	81.75	8.50	19.2
Highest Wingspan Advantage	Kelly Oubre Jr.	SF	77.75	8.50	21.5

The table below shows the averages for each group. The high wingspan advantage group has a higher average for some defensive stats, especially blocks. This shows a possible pattern, but not a perfect one.

Table 4: Average Stats for Wingspan Advantage Extremes

Group	Players	Mean Wingspan Advantage	Mean PTS	Mean BLK	Mean TRB
Highest Wingspan Advantage	10	9.22	23.78	1.51	10.52
Lowest Wingspan Advantage	10	-0.70	24.22	0.72	10.36

The scatter plot below compares wingspan advantage and blocks. There is a small upward trend, but position still matters. Centers and forwards usually get more block chances because of where they play.

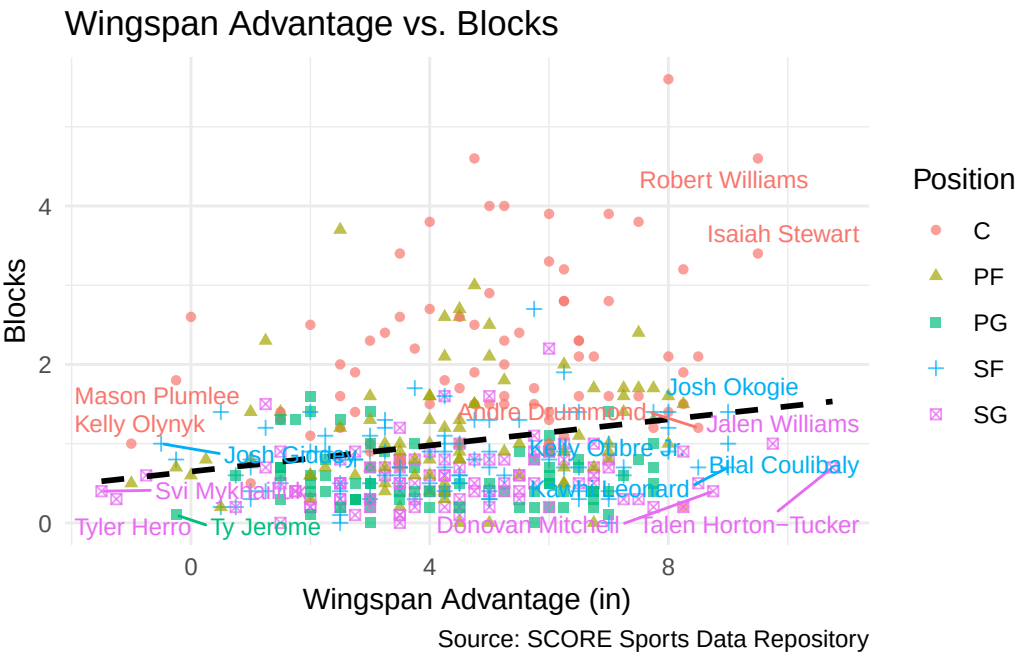


Figure 1: Wingspan Advantage Compared to Blocks by Position

Alt text: Scatter plot of wingspan advantage and blocks. Points are colored and shaped by position, and the highest and lowest wingspan advantage players are labeled. The scatter plot below compares wingspan advantage and total rebounds. This relationship is less clear than blocks. Some players with lower wingspan advantage still rebound well, so rebounding depends on more than arm length.

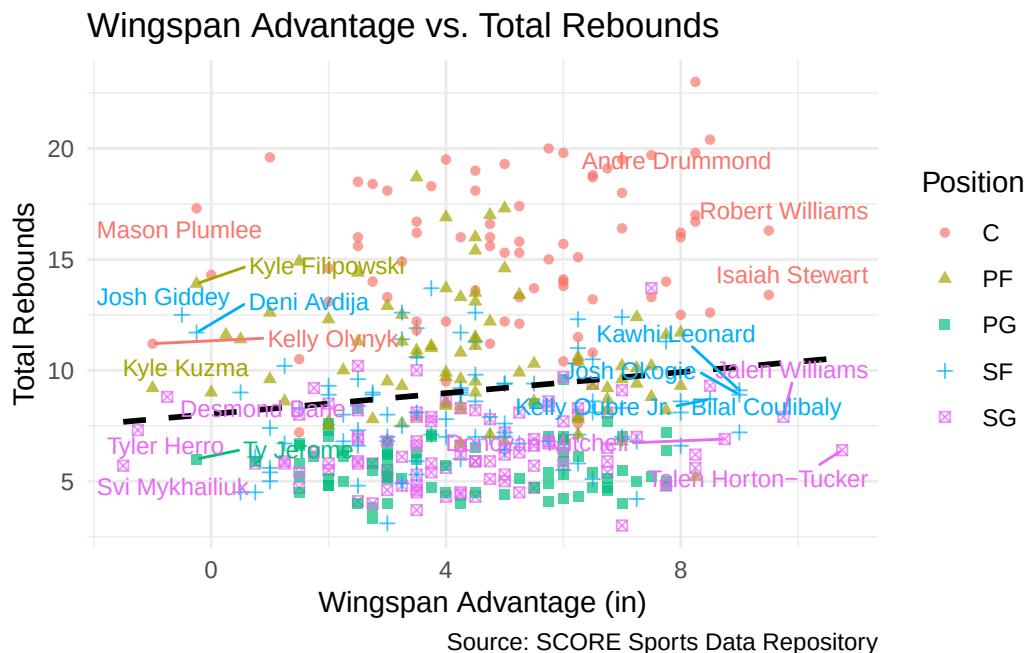


Figure 2: Wingspan Advantage Compared to Total Rebounds by Position

Alt text: Scatter plot of wingspan advantage and rebounds. Points are colored and shaped by position, and the highest and lowest wingspan advantage players are labeled.

Overall, higher wingspan advantage seems connected to better defensive stats, especially blocks. However, position still matters a lot. Wingspan advantage is useful to look at, but it does not explain performance by itself.

6 Summary

From the visualizations and tables, we now have a better understanding of how physical characteristics give basketball players an advantage on the court. We can see how wingspan specifically increases defensive statistics such as blocks, but position on the court still matters a lot. This makes sense though from the first visualization we understand how positions on the court are also determined by physical characteristics. We saw how certain positions have on average a larger height and wingspan. From this we can see how large of an impact physical characteristics have on basketball performance. A taller player with a larger wingspan is more likely to have better defensive stats and play in a position to perform better. Although there are other factors to consider such as skill and age, we can see a direct relationship through our visuals between physical characteristics and defensive performance.

7 Author Contributions

For this assignment Sean Kennelly created the README.md file, the project plan, and the introduction in the qmd. He also worked on the first visualization and analysis plot. Charlie Coyle worked on the second visualization and analysis. Suhas Rakothu worked on the third visualization and analysis.

8 Code Appendix

This appendix contains all R code used to import, clean, analyze, and visualize the data for our dataset

8.1 Physical Characteristics by Position Summary Table

```
# Primary Author: Sean
# Reviewed by: Suhas

library(tidyverse)
library(knitr)
library(kableExtra)

nba_data <- read_csv("nba_wingspan_performance_2025.csv")
```

```
Rows: 419 Columns: 26
-- Column specification -----
Delimiter: ","
chr  (3): name, team, position
dbl (23): height_inches, wingspan_inches, wingspan_advantage, g, mp, pts, or...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
nba_data |>
  group_by(position) |>
  summarize(
    Mean_Height = round(mean(height_inches, na.rm = TRUE), 2),
    SD_Height = round(sd(height_inches, na.rm = TRUE), 2),
    Mean_Wingspan = round(mean(wingspan_inches, na.rm = TRUE), 2),
    SD_Wingspan = round(sd(wingspan_inches, na.rm = TRUE), 2),
    Mean_Advantage = round(mean(wingspan_advantage, na.rm = TRUE), 2),
    SD_Advantage = round(sd(wingspan_advantage, na.rm = TRUE), 2)
```



```

) |>
kable(
  caption = "Physical Characteristics by Position",
  col.names = c("Position", "Mean", "SD", "Mean", "SD", "Mean", "SD")
) |>
add_header_above(c(" " = 1, "Height (in)" = 2, "Wingspan (in)" = 2, "Wingspan Advantage (in)" = 1))
kable_styling(bootstrap_options = "striped")

```

Table 5: Physical Characteristics by Position

Position	Height (in)		Wingspan (in)		Wingspan Advantage (in)	
	Mean	SD	Mean	SD	Mean	SD
C	82.31	1.79	87.38	2.64	5.08	2.29
PF	80.02	1.66	84.42	1.95	4.40	2.10
PG	74.05	1.64	78.33	2.51	4.28	2.09
SF	78.26	1.32	82.49	2.09	4.24	2.31
SG	76.23	1.37	80.58	2.22	4.35	2.20

8.2 Physical Characteristics by Position Box Plots

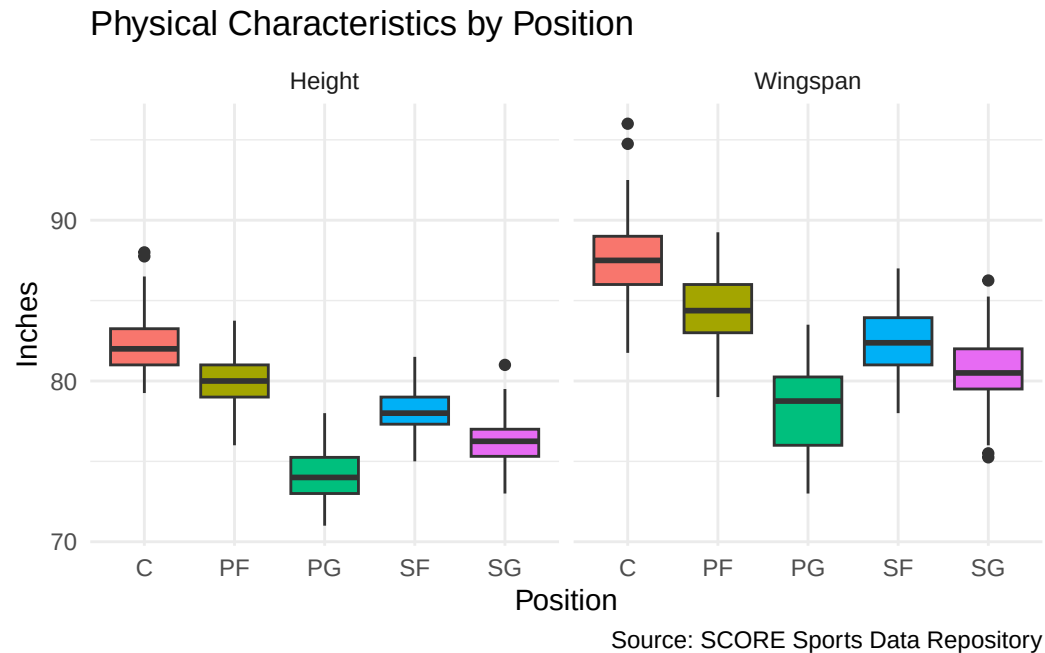
```

# Primary Author: Sean
# Reviewed by: Suhas

nba_data |>
  select(position, height_inches, wingspan_inches) |>
  pivot_longer(
    cols = c(height_inches, wingspan_inches),
    names_to = "measurement",
    values_to = "inches"
  ) |>
  mutate(
    measurement = recode(measurement,
                          "height_inches" = "Height",
                          "wingspan_inches" = "Wingspan"
    )
  ) |>
  ggplot(aes(x = position, y = inches, fill = position)) +
  geom_boxplot() +
  facet_wrap(~measurement) +
  labs(
    title = "Physical Characteristics by Position",
    x = "Position",
    y = "Inches",
    fill = "Position",
  )

```

```
caption = "Source: SCORE Sports Data Repository"
) +
theme_minimal() +
theme(legend.position = "none")
```



8.3 Wingspan Analysis Table Code

```
# Primary Author: Charlie
# Reviewed by: Sean

# --- Load libraries ---
library(dplyr)
library(readr)
library(knitr)

# --- Load data ---
# Replace with your link
df <- read_csv("https://data.scorenetwork.org/data/nba_wingspan_performance_2025.csv")
```

Rows: 419 Columns: 26

-- Column specification -----

Delimiter: ","

chr (3): name, team, position

dbl (23): height_inches, wingspan_inches, wingspan_advantage, g, mp, pts, or...

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
# --- Create wingspan advantage tiers ---
df <- df %>%
  mutate(
    wingspan_tier = case_when(
      wingspan_advantage < 4 ~ 'Low (<4")',
      wingspan_advantage >= 4 & wingspan_advantage <= 6 ~ 'Medium (4-6")',
      wingspan_advantage > 6 ~ 'High (>6")'
    )
  )

# --- Group and compute averages ---
summary_table <- df %>%
  group_by(wingspan_tier) %>%
  summarise(
    avg_blk = mean(blk, na.rm = TRUE),
    avg_trb = mean(trb, na.rm = TRUE),
    avg_stl = mean(stl, na.rm = TRUE),
    avg_pts = mean(pts, na.rm = TRUE)
  )

# --- View result ---
kable(summary_table)
```

Table 6

wingspan_tier	avg_blk	avg_trb	avg_stl	avg_pts
High (>6")	1.2276190	9.709524	1.766667	21.78952
Low (<4")	0.7958824	8.381177	1.613529	21.74235
Medium (4-6")	1.1209790	9.454546	1.743357	21.17622

8.3.1 Wingspan Analysis visualization Code

```
# Primary Author: Charlie
# Reviewed by: Sean
library(dplyr)
library(tidyr)
library(ggplot2)

df <- read.csv("https://data.scorenetwork.org/data/nba_wingspan_performance_2025.csv")

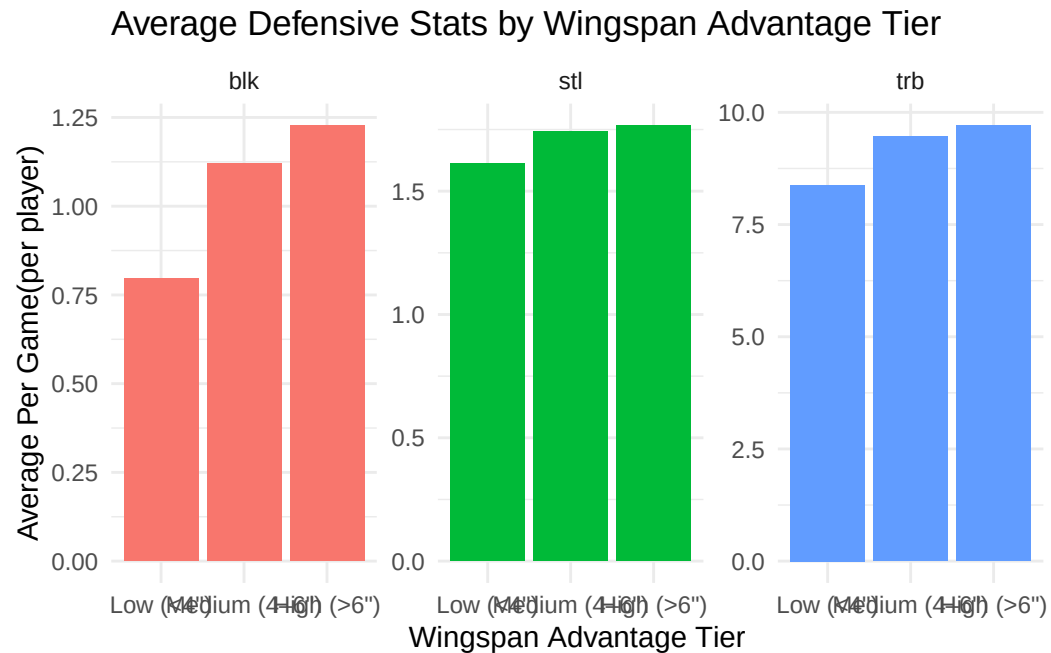
summary_long <- df %>%
```

```

mutate(
  wingspan_tier = case_when(
    wingspan_advantage < 4 ~ 'Low (<4)"',
    wingspan_advantage >= 4 & wingspan_advantage <= 6 ~ 'Medium (4-6)"',
    wingspan_advantage > 6 ~ 'High (>6)"'
  ),
  wingspan_tier = factor(
    wingspan_tier,
    levels = c('Low (<4)"', 'Medium (4-6)"', 'High (>6)"')
  )
) %>%
group_by(wingspan_tier) %>%
summarise(
  blk = mean(blk, na.rm = TRUE),
  trb = mean(trb, na.rm = TRUE),
  stl = mean(stl, na.rm = TRUE)
) %>%
pivot_longer(
  cols = c(blk, trb, stl),
  names_to = "stat",
  values_to = "average"
)

ggplot(summary_long, aes(x = wingspan_tier, y = average, fill = stat)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ stat, scales = "free_y") +
  labs(
    title = "Average Defensive Stats by Wingspan Advantage Tier",
    x = "Wingspan Advantage Tier",
    y = "Average Per Game(per player)"
  ) +
  theme_minimal()

```



8.4 Suhas Wingspan Extremes Table Code

```
# Primary Author: Suhas
# Reviewed by: Charlie

library(tidyverse)
library(knitr)
library(kableExtra)
library(ggrepel)

nba_data <- read_csv("nba_wingspan_performance_2025.csv")

players_clean <- nba_data |>
  select(
    name,
    position,
    height_inches,
    wingspan_advantage,
    pts,
    blk,
    trb,
    stl
  ) |>
  filter(
    !is.na(name),
    !is.na(position),
```

```

    !is.na(height_inches),
    !is.na(wingspan_advantage),
    !is.na(pts),
    !is.na(blk),
    !is.na(trb),
    !is.na(stl)
  )

lowest_wingspan <- players_clean |>
  arrange(wingspan_advantage) |>
  slice_head(n = 10) |>
  mutate(group = "Lowest Wingspan Advantage")

highest_wingspan <- players_clean |>
  arrange(desc(wingspan_advantage)) |>
  slice_head(n = 10) |>
  mutate(group = "Highest Wingspan Advantage")

wingspan_extremes <- bind_rows(
  lowest_wingspan,
  highest_wingspan
)

wingspan_extremes |>
  select(
    group,
    name,
    position,
    height_inches,
    wingspan_advantage,
    pts,
    blk,
    trb,
    stl
  ) |>
  mutate(
    height_inches = round(height_inches, 2),
    wingspan_advantage = round(wingspan_advantage, 2),
    pts = round(pts, 2),
    blk = round(blk, 2),
    trb = round(trb, 2),
    stl = round(stl, 2)
  ) |>
  kable(
    caption = "Players with the Highest and Lowest Wingspan Advantage",
    col.names = c(
      "Group",

```

```

    "Player",
    "Position",
    "Height (in)",
    "Wingspan Advantage (in)",
    "PTS",
    "BLK",
    "TRB",
    "STL"
  )
) |>
kable_styling(bootstrap_options = "striped")

```

Table 7: Players with the Highest and Lowest Wingspan Advantage

Group	Player	Position	Height (in)	Wingspan Advantage (in)	PTS
Lowest Wingspan Advantage	Svi Mykhailiuk	SG	78.50	-1.50	21.1
Lowest Wingspan Advantage	Tyler Herro	SG	76.50	-1.25	33.7
Lowest Wingspan Advantage	Kelly Olynyk	C	82.75	-1.00	20.8
Lowest Wingspan Advantage	Kyle Kuzma	PF	80.00	-1.00	23.9
Lowest Wingspan Advantage	Desmond Bane	SG	77.00	-0.75	28.0
Lowest Wingspan Advantage	Josh Giddey	SF	80.00	-0.50	22.5
Lowest Wingspan Advantage	Deni Avdija	SF	80.75	-0.25	27.4
Lowest Wingspan Advantage	Kyle Filipowski	PF	82.75	-0.25	21.8
Lowest Wingspan Advantage	Mason Plumlee	C	83.25	-0.25	12.7
Lowest Wingspan Advantage	Ty Jerome	PG	76.25	-0.25	30.3
Highest Wingspan Advantage	Talen Horton-Tucker	SG	74.50	10.75	24.4
Highest Wingspan Advantage	Jalen Williams	SG	76.50	9.75	32.0
Highest Wingspan Advantage	Isaiah Stewart	C	79.25	9.50	14.6
Highest Wingspan Advantage	Robert Williams	C	80.00	9.50	16.1
Highest Wingspan Advantage	Bilal Coulibaly	SF	78.00	9.00	17.8
Highest Wingspan Advantage	Josh Okogie	SF	75.00	9.00	22.3
Highest Wingspan Advantage	Kawhi Leonard	SF	78.00	9.00	33.2
Highest Wingspan Advantage	Donovan Mitchell	SG	73.25	8.75	36.7
Highest Wingspan Advantage	Andre Drummond	C	81.75	8.50	19.2
Highest Wingspan Advantage	Kelly Oubre Jr.	SF	77.75	8.50	21.5

```

wingspan_extremes |>
  group_by(group) |>
  summarize(
    Players = n(),
    Mean_Wingspan_Advantage = round(mean(wingspan_advantage), 2),
    Mean_PTS = round(mean(pts), 2),
    Mean_BLK = round(mean(blk), 2),
    Mean_TRB = round(mean(trb), 2),

```

```

    Mean_STL = round(mean(stl), 2)
  ) |>
  kable(
    caption = "Average Stats for Wingspan Advantage Extremes",
    col.names = c(
      "Group",
      "Players",
      "Mean Wingspan Advantage",
      "Mean PTS",
      "Mean BLK",
      "Mean TRB",
      "Mean STL"
    )
  ) |>
  kable_styling(bootstrap_options = "striped")

```

Table 8: Average Stats for Wingspan Advantage Extremes

Group	Players	Mean Wingspan Advantage	Mean PTS	Mean BLK	Mean TRB
Highest Wingspan Advantage	10	9.22	23.78	1.51	10.52
Lowest Wingspan Advantage	10	-0.70	24.22	0.72	10.36

8.4.1 Suhas Wingspan Extremes Visualization Code

```

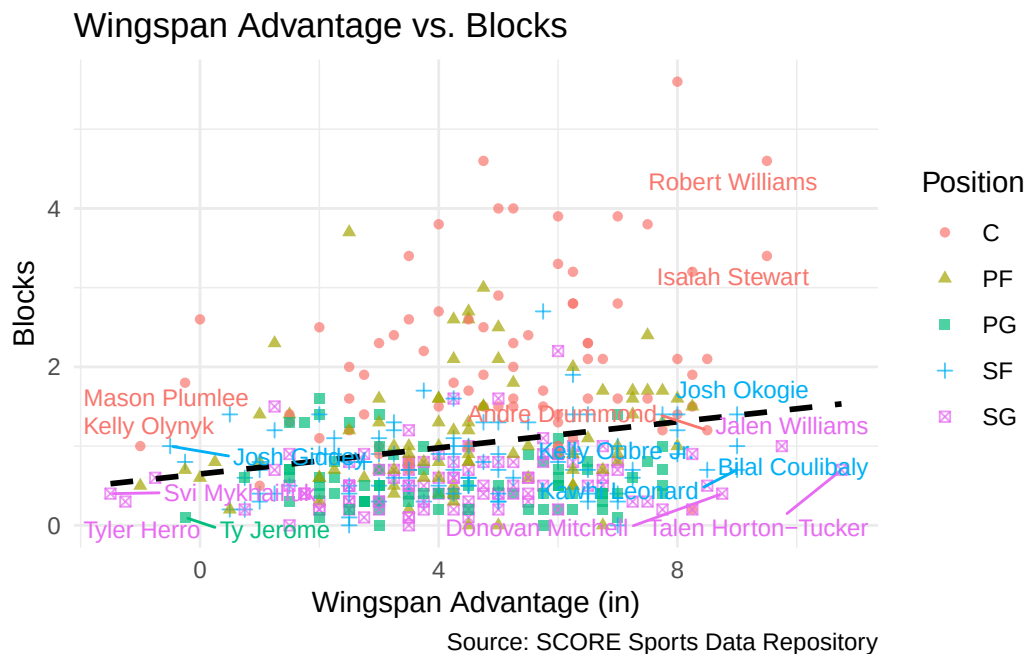
# Primary Author: Suhas
# Reviewed by: Charlie

ggplot(
  players_clean,
  aes(
    x = wingspan_advantage,
    y = blk,
    color = position,
    shape = position
  )
) +
  geom_point(alpha = 0.7) +
  geom_smooth(
    aes(group = 1),
    method = "lm",
    se = FALSE,
    color = "black",
    linetype = "dashed"
  ) +

```



```
geom_text_repel(
  data = wingspan_extremes,
  aes(label = name),
  size = 3,
  show.legend = FALSE
) +
labs(
  title = "Wingspan Advantage vs. Blocks",
  x = "Wingspan Advantage (in)",
  y = "Blocks",
  color = "Position",
  shape = "Position",
  caption = "Source: SCORE Sports Data Repository"
) +
theme_minimal()
```



```
ggplot(
  players_clean,
  aes(
    x = wingspan_advantage,
    y = trb,
    color = position,
    shape = position
  )
) +
geom_point(alpha = 0.7) +
geom_smooth(
```

```

aes(group = 1),
method = "lm",
se = FALSE,
color = "black",
linetype = "dashed"
) +
geom_text_repel(
  data = wingspan_extremes,
  aes(label = name),
  size = 3,
  show.legend = FALSE
) +
labs(
  title = "Wingspan Advantage vs. Total Rebounds",
  x = "Wingspan Advantage (in)",
  y = "Total Rebounds",
  color = "Position",
  shape = "Position",
  caption = "Source: SCORE Sports Data Repository"
) +
theme_minimal()

```

